Functional Description

Gas engine-generator sets

GG06R0400A1

with engines B3066L8, E3066L9/LH9 and B3066L9

MS13076/00E

Name	SLU name Model	Engine model	Application group
MTU 06R400 GS	GG06R0400A1	B3066L8	3A, continuous operation, unrestricted
MTU 06R400 GS	GG06R0400A1	E3066L9	3A, continuous operation, unrestricted
MTU 06R400 GS	GG06R0400A1	E3066LH9	3A, continuous operation, unrestricted
MTU 06R400 GS	GG06R0400A1	B3066L9	3A, continuous operation, unrestricted

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1 Engine-Generator Set

1.1 Application group

The genset consists of a gas engine and a generator mounted on a common baseframe. Upon demand, the engine will start and drives the generator to generate heat energy and to produce electrical power.

There are two options for dissipating heat:
- Heat dissipation at the genset
- Heat dissipation provided by the customer

The object of delivery from MTU Onsite Energy complies with the EC Machinery Directive and German specifications and standards.

When the object of delivery is used abroad or in other special regions, MTU Onsite Energy is not responsible for observance of the statutory and other specifications / standards applicable at the place of use.

Application group 3A – Continuous operation, unrestricted (Continuous Power)

Gensets for unrestricted continuous power and heat energy supply are used for:
- Base load generation
- Part of a power supply or the public grid

Continuous operation	Application group 3A
Operating mode	Continuous operation, unrestricted
Rating definition	10% overload capacity (ICXN)
Load factor	< 100 %
Operating hours	Unrestricted

Benefits

- Wide range of standardized gensets to meet customer requirements with regard to power, emission and other characteristics.
- Option of selecting different components:
  - Only current supply
  - Heat supply from cogeneration from engine coolant
  - Heat supply from cogeneration from engine coolant and exhaust gas
- Most recent gas engine technology
- Components are designed with state-of-the-art technology for high electrical and thermal efficiency and long service life.

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1.2 Nameplate and designation

Nameplates

The genset has the following nameplates:
- Nameplate of genset
- Engine nameplate
- Nameplate of generator

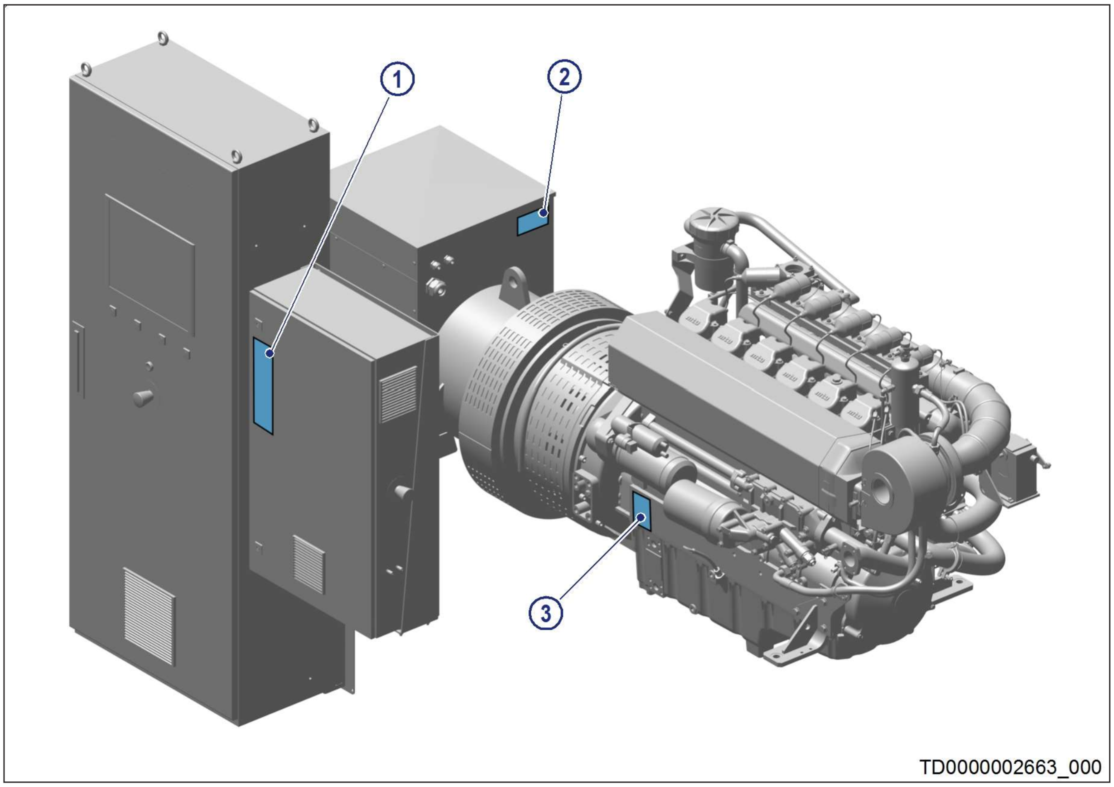

Figure 1: Fitting position of nameplates

1 Nameplate of genset
2 Nameplate of generator
3 Engine nameplate

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Genset – nameplate and designation

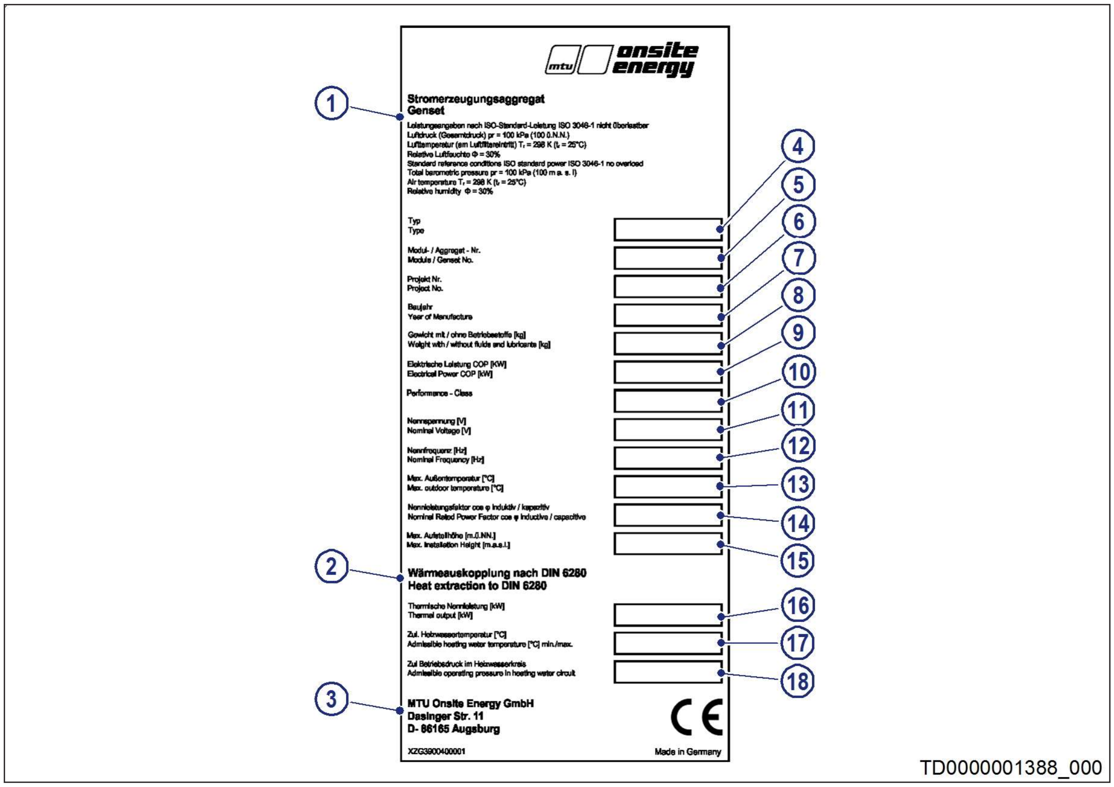

Figure 2: Nameplate of genset

Item	Description
1	Power ratings as per ISO standard power ISO 3046-1 no overload capability Air pressure (total pressure) pr = 100 kPa (100 above sea level) Air temperature (at air filter inlet) T_r = 298 K (t_r = 25 °C) Relative air humidity ϕ = 30%
2	Heat supply from cogeneration as per DIN 6280
3	MTU Onsite Energy Dasinger Str. 11 D-86165 Augsburg
4	Type
5	Module / genset no.
6	Project no.
7	Year of construction
8	Weight with / without fluids and lubricants [kg]
9	Electrical output COP [KW]
10	Performance-Class (performance class of generator)
11	Rated voltage [V]
12	Rated frequency [Hz]
13	Max. outside temperature [°C]
14	Rated power factor cos ϕ inductive / capacitive
15	Max. site altitude [m above sea level]
16	Thermal rated power [kW]

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Item	Description
17	Perm. heating water temperature [°C]
18	Perm. operating pressure in heating water circuit

Explanation of type

Example:MTU 6R400 GS	
MTU	Manufacturer: MTU Onsite Energy
6	Number of cylinders of the engine
R	Cylinder arrangement
400	400 series
GS	Gas system

Explanation of model name (SLU name)

Example:GG06R0400A1	
GG	Product type: Gasgenset
06	Number of cylinders of the engine
R	Cylinder arrangement
0400	400 series
A1	Design/version

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1.3 Module – With heat recovery and integrated exhaust gas heat exchanger

Description

The module comprises the following main components:
- Engine
- Gas train (not visible on the illustrations)
- Generator and coupling
- Heat recovery with exhaust gas heat exchanger
- MMC (MTU Module Control) for plant control, closed-loop control, diagnostics
- MIP (MTU Interface Panel) for engine control, closed-loop control, diagnostics

The energy contained in the fuel is converted into mechanical work and heat in the combustion engine.

The generator, which converts the mechanical work into electrical energy, is flange-mounted on the engine via a resilient coupling. The bell housing is provided with a service port that permits the replacement of the flexible coupling element without displacing the engine or generator. Engine and generator are attached with resilient, vibration-damping elements to the baseframe.

The heat is transferred via the coolant at several points:
- Lube oil heat
- Engine heat
- Exhaust gas heat

The plate-core heat exchanger represents the interface between the coolant system of the module and the customer-provided heating system. In this connection, it is also referred to as a coolant or heating water heat exchanger.

There are two types of mixture cooling:
- **External mixture cooling**
  Mixture coolant dissipates the heat at the mixture cooler via a customer-provided (external) cooler.
- **Internal mixture cooling**
  Customer-provided heating water dissipates the heat at the mixture cooler. This heating water (internal) is then directed via the plate-core heat exchanger.

All closed-loop controls, control systems and monitoring systems, including communication options to and from the system, are implemented in the hardware and software in the control cabinets.

The following components have complete pipework and are installed in the baseframe:
- Plate-core heat exchanger
- Exhaust gas heat exchanger
- Silencer
- Catalytic converter (optional)

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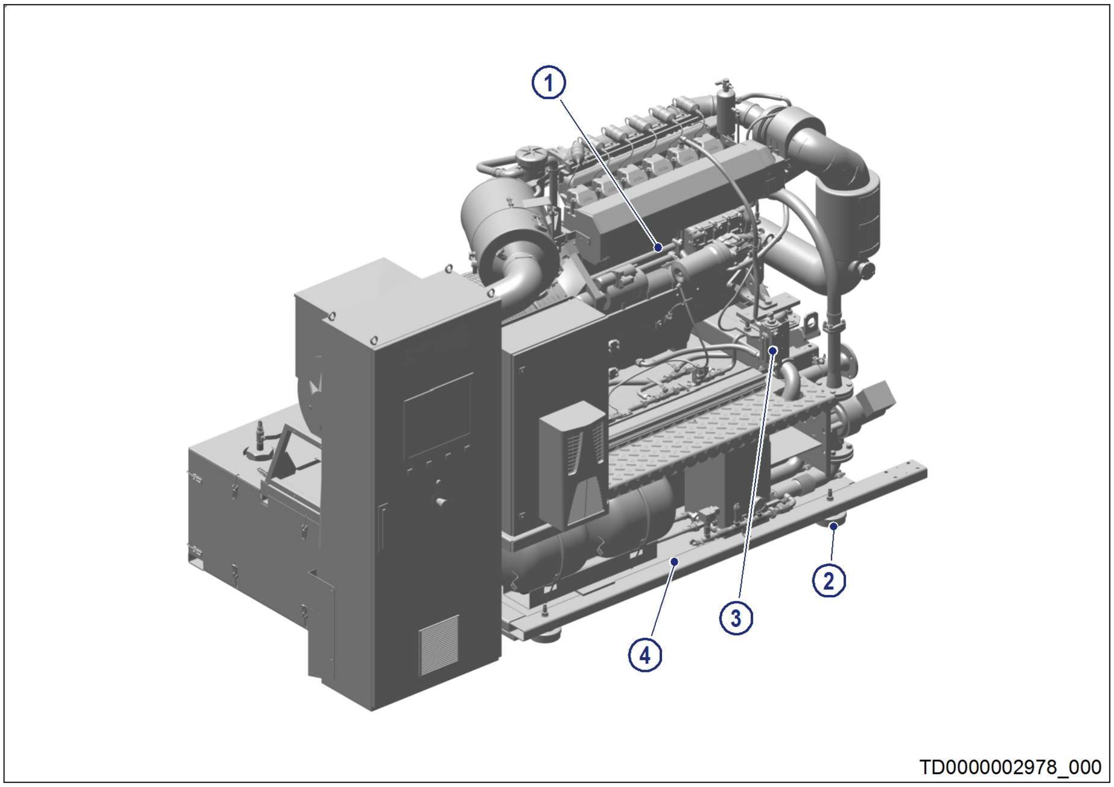

Figure 3: Module, generator side

1	Engine	3	Oil sight glass
2	Machine shoe (optional)	4	Baseframe

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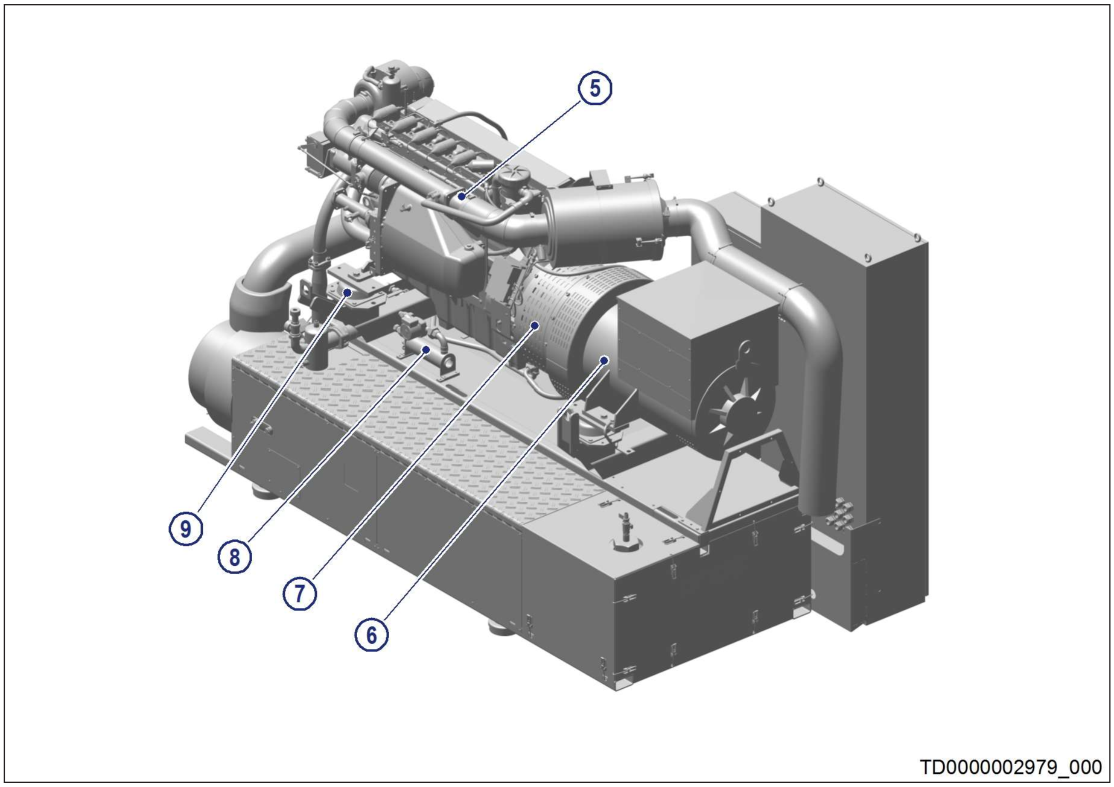

Figure 4: Module, generator side

5 Gas mixer (gas train connection)
6 Generator
7 Flange coupling
8 Anti-condensation heating (optional)
9 Bowl-shaped element

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